

**ELEC 477 Operating Systems**

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Exam #3

Time: 25 Minutes

**Instructions:** Total 19 marks. Closed Book.  
Answer all questions in the space provided on these pages  
If you need extra space, use the extra pages at the end of the exam.  
Make sure you indicate to the marker you are using the extra space.

1. (4 marks) Give a brief (one or two line) description of each of the following terms as they were discussed in class.

a. Omission Failure

Fails to take an action that should have even taken (or fails to respond to a request)  
8C2 slide 6-7

b. Event Driven System

system responds (reacts) to external events (9C2 Slide 9)

c. Push Based Update

When one replica is updated, the update is pushed to all of the other replicas (7C3, slide 23)

d. Renewal frequency lease

If a replica does a pull every time the lease expires, then the lease isn't long enough and must be increased (7C3, slide 35)

2. (3 marks) How do the type of failures determine how many servers there must be for the group to be  $k$ -fault tolerant?

If the failure can be detected (Fail stop (crash/ timeout) then  $k+1$

if the failure is arbitrary (e.g. still responding, producing incorrect results) then  $2k+1$

(8C2 slide 13)

3.(4 marks) One of the ways for servers to initiate replicas of data items is to count accesses to each data item. What are the three actions that a server can take on a replicated data item, and how does the server decide?

It can replicate the item to another server, if enough requests (threshold  $R$ ) come from clients close to that server

It can delete the current replica (assuming it exists elsewhere) if there are not enough accesses (threshold  $D$ ) from clients close to the current replica.

Migration threshold. If a large enough proportion of client access come from clients closer to another server(threshold  $M$ ), the replica might be moved to another server.

(7C3 slide 17-28)

4. (4 marks) In Quorum based replica consistency protocols with  $N$  servers, there are two thresholds,  $N_W$  and  $N_R$ . (8C1 slides 4-7)

a. (2 marks) What is the lower bound on the value of  $N_W$ ? Why?

the lower bound of  $N_W$  is  $N/2+1$ . This ensures that there are always a majority of replicas with the same value. Otherwise there could be two sets with a new version number that conflict.

b (2 marks) What is the lower bound on the value of  $N_R$ ? What happens if the value is lower?

The lower bound is  $N_R > N - N_W$  ( $N_R + N_W > N$ ). This ensures that a copy of the latest value is found in one of values that was read.

5 (2 marks). In Client Consistency protocols each client keeps track of two sets of identifiers? How are they identifiers created, and what are the two sets?

(8C1 slide 8) Each time a data replica is written, a globally unique identifier is created for the write operation.

The write set is the last write ids of each of the replicas that were written by the client.

The read set is the last write id of each of the replicas that was read by the client.

6. (2 marks) The key data structure in the Raft Fault Tolerance Algorithm is the log. What data is kept in each entry in the log?

Each entry in the log contains the operation and the term number. 8C3 slide 16

Extra Space.

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